Question 1

1. Consider the observation model $x_n = \log\theta + w_n \quad n = 1,\ldots,N$, where $w_n \sim N(0,1)$ are i.i.d. and $\theta > 0$ is an unknown deterministic parameter.

a) Find Estimator

a. Propose an estimator of θ from the observations $\mathbf{x}^{(N)} = [x_1,\ldots,x_N]^T$.

Since θ is a deterministic parameter we will prefer using ML estimator.

- We will derive the expression of the ML estimator by definition :

$$\hat{\theta}_{ML}(\underline{x}) = \underset{\theta\in\Omega_\theta}{argmax}\left\{f_{\underline{x}|\theta}(\underline{x}|\theta)\right\}$$

- For that purpose, we will find the conditional distribution of $\underline{x}|\theta$:

$$x_n = \log\theta + w_n : w_n \overset{i.i.d}{\sim} \mathcal{N}(0,1)$$

$$\underline{x}^{(N)} \sim \mathcal{N}(\mathbb{1}_N\cdot\log\theta, I_N) \;:\; \mathbb{1}_N = \begin{pmatrix}1\\1\\\vdots\\1\end{pmatrix} \in \mathbb{R}^N$$

- We can derive the conditional joint PDF of the samples :

$$f_{\underline{x}|\theta}(\underline{\alpha};\theta) = \prod_{i=1}^{N} f_{x_i|\theta}(\alpha_i;\theta) = \prod_{i=1}^{N}\frac{1}{\sqrt{2\pi}}\cdot e^{-\frac{(x_i-\log\theta)^2}{2}} = (2\pi)^{-\frac{N}{2}}\cdot\exp\left[-\frac{1}{2}\sum_{i=1}^{N}(x_i-\log\theta)^2\right]$$

- We notice that for convenience, we can find the max argument of the $\ln$ of the conditional joint distribution without changing our goal :

$$\hat{\theta}_{ML}(\underline{x}) = \underset{\theta\in\Omega_\theta}{argmax}\left\{f_{\underline{x}|\theta}(\underline{x}|\theta)\right\} = \underset{\theta\in\Omega_\theta}{argmax}\left\{\ln f_{\underline{x}|\theta}(\underline{x}|\theta)\right\}$$

$$\cdots = \ln\left[(2\pi)^{-\frac{N}{2}}\cdot\exp\left[-\frac{1}{2}\sum_{i=1}^{N}(x_i-\log\theta)^2\right]\right] = -\frac{N}{2}\ln(2\pi) + \left[-\frac{1}{2}\sum_{i=1}^{N}(x_i-\log\theta)^2\right]$$

- We will find the max argument by taking the derivative and comparing to 0 :

$$\xrightarrow{\frac{d}{d\theta}} -\cancel{\frac{1}{2}}\sum_{i=1}^{N}\cancel{2}\cdot(x_i-\log\theta)\cdot\left(-\frac{1}{\theta}\right) = -\cancel{\frac{1}{\theta}}\left[N\cdot\log\theta - \sum_{i=1}^{N}x_i\right] = 0$$

$$\theta > 0 \implies N \log \hat{\theta} = \sum_{i=1}^{N} x_i \implies \hat{\theta} = \exp\left[\frac{1}{N} \cdot \sum_{i=1}^{N}\right]$$

✓ **Result**

We get the final derivation of the MLE :

$$\hat{\theta}_{ML}\left(\underline{x}^{(N)}\right) = \exp\left[\frac{1}{N} \cdot \sum_{i=1}^{N}\right] = \exp(\bar{x})$$

b) Analyzing The Results

b. Using MATLAB/ Python, evaluate the bias, variance, and MSE of the estimates for different values of N and θ as follows. Let $\hat{\theta}_N = \hat{\theta}\left(\mathbf{x}^{(N)}\right)$ denote the proposed estimator and define the estimation error for the i^{th} realization of $\mathbf{x}^{(N)}$ as $\varepsilon_N\left(\mathbf{x}_i^{(N)}, \theta\right) = \hat{\theta}\left(\mathbf{x}_i^{(N)}\right) - \theta$ where $i = 1, \ldots, L$ stands for an index of the i^{th} realization of $\mathbf{x}^{(N)}$. The bias, variance, and MSE of the estimates are given by $bias(N,\theta) = \frac{1}{L}\sum_{i=1}^{L} \varepsilon_N\left(\mathbf{x}_i^{(N)}, \theta\right)$, $\quad \mathrm{var}(N,\theta) = \frac{1}{L}\sum_{i=1}^{L}\left(\varepsilon_N\left(\mathbf{x}_i^{(N)}, \theta\right) - bias(N,\theta)\right)^2$

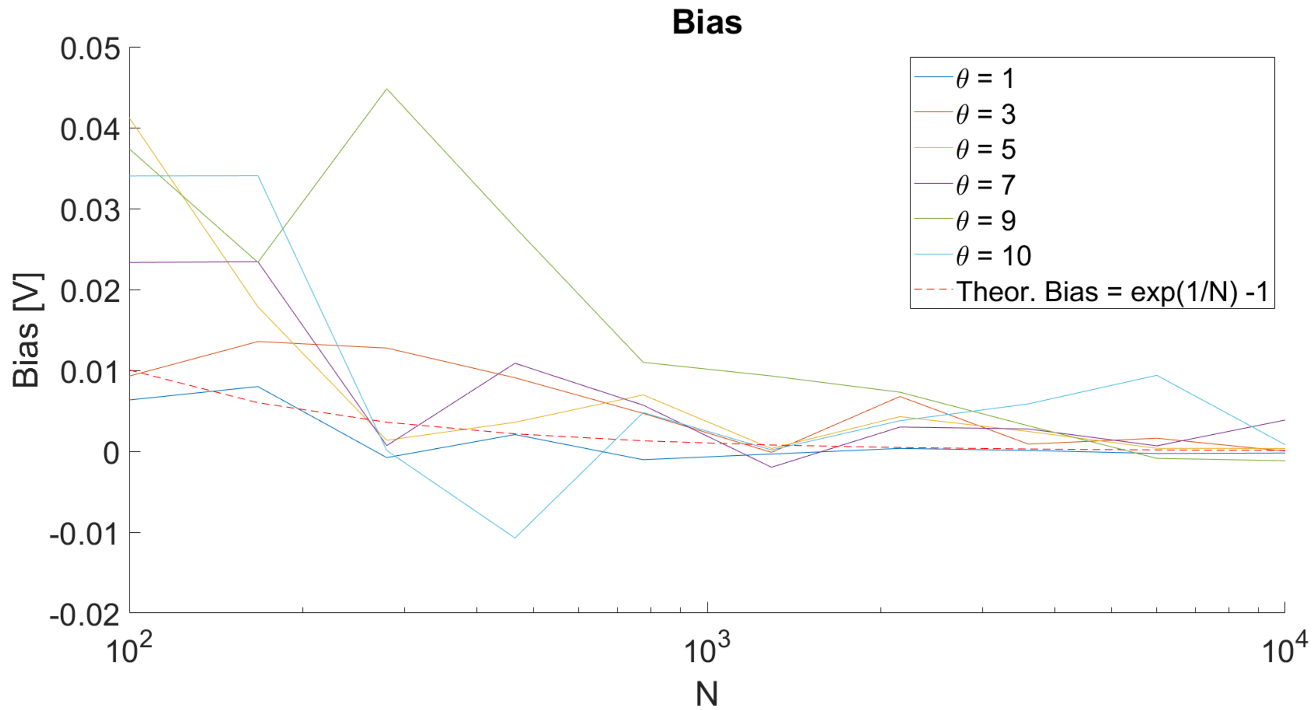
$\mathrm{MSE}(N,\theta) = \frac{1}{L}\sum_{i=1}^{L} \varepsilon_N^2\left(\mathbf{x}_i^{(N)}, \theta\right)$, respectively, and $L = 1000$.

i. Plot the bias, variance, MSE of the estimator and the Cramér-Rao bound (CRB) for estimation of θ, versus N for at least 5 different values of N. In each figure, include plots for 6 different values of θ in the interval $0.1 \le \theta \le 10$ where the N axis is logarithmic. In the figures of the variance, and MSE use a logarithmic scale for both axes.

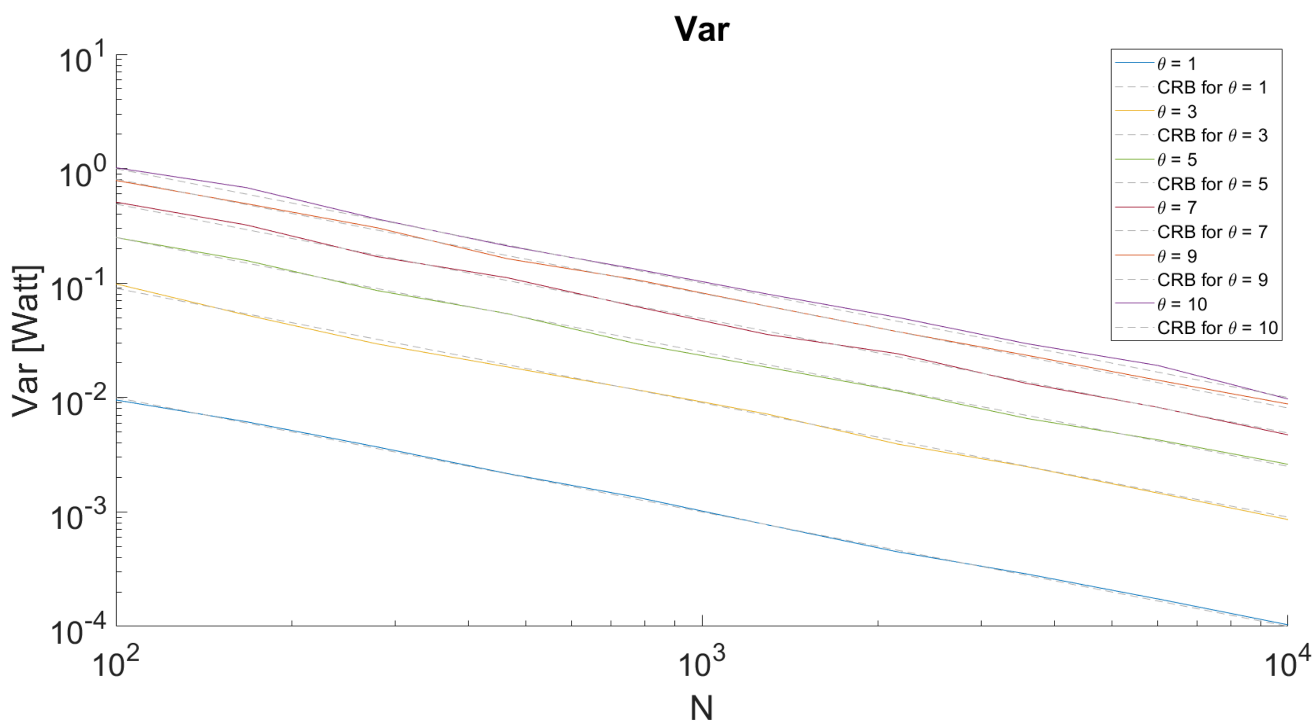
The results of said simulation are as follows

Plotted as a function of N :

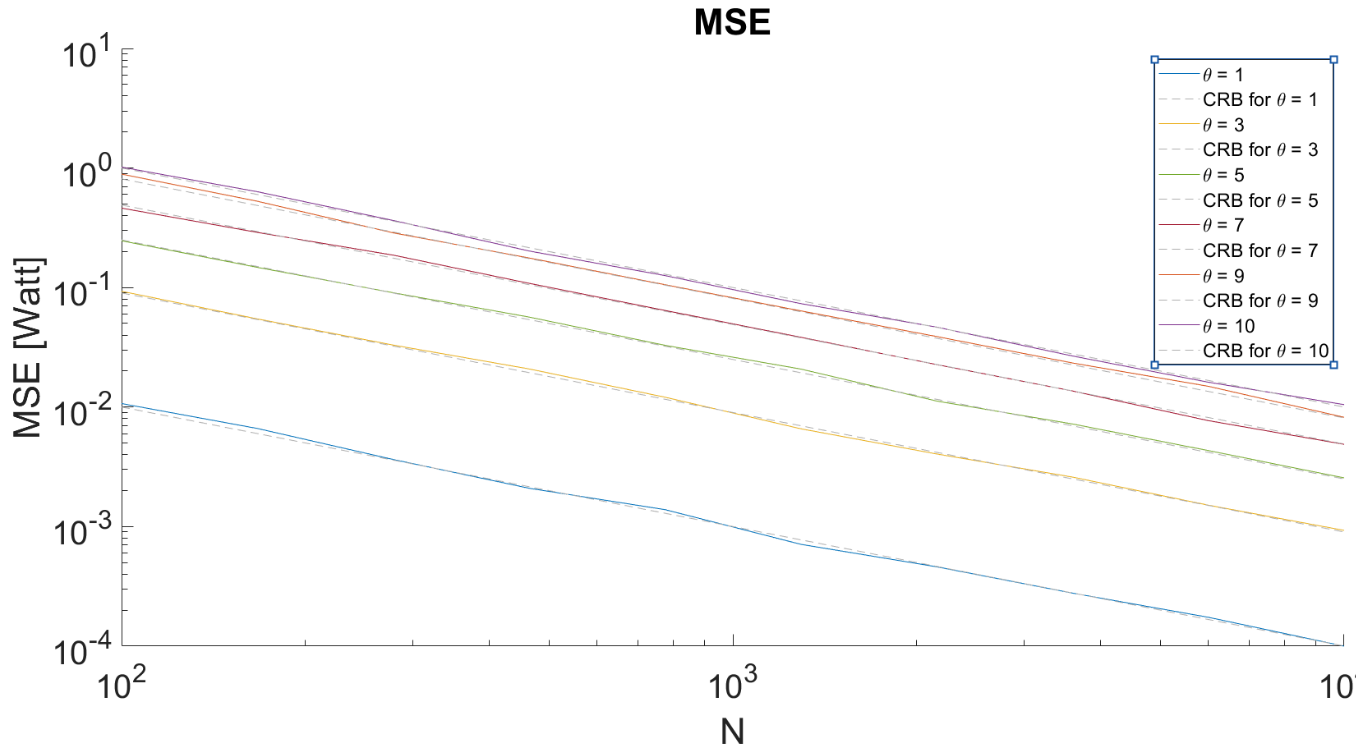
$Bias(N, \theta_i)$:



$Var(N, \theta_i)$:

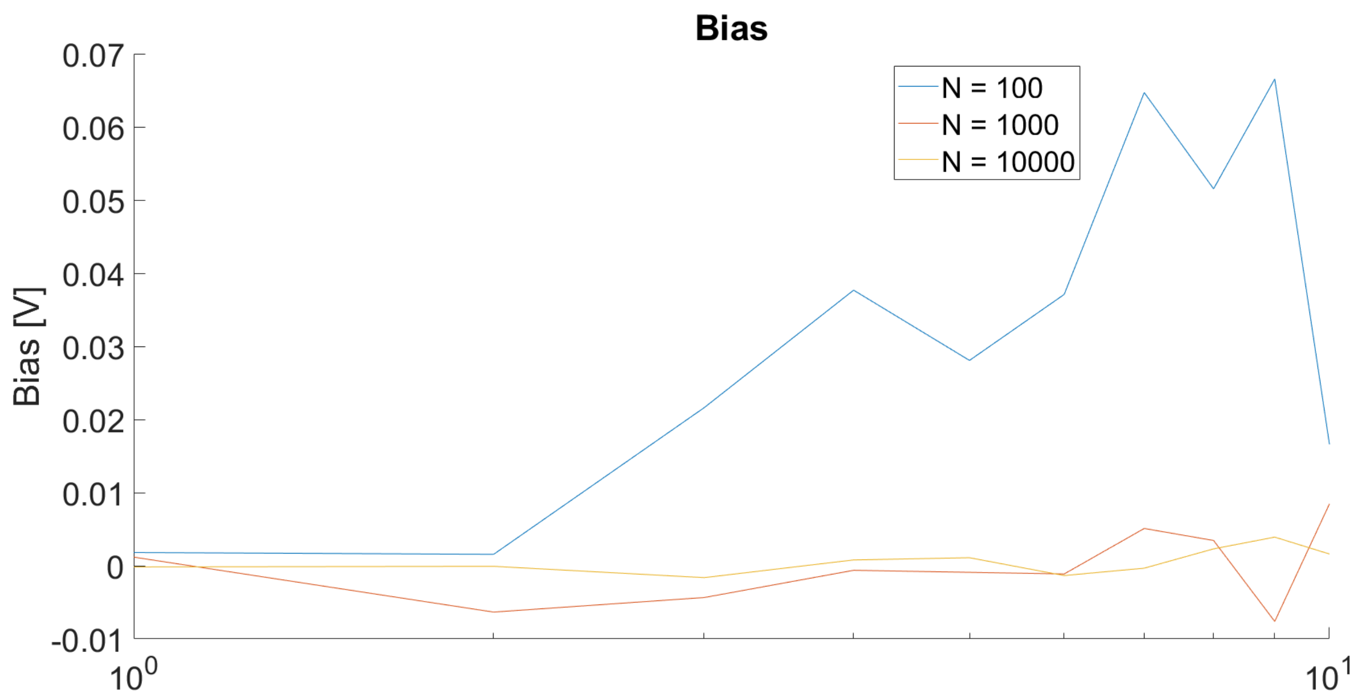


$MSE(N, \theta_i)$:

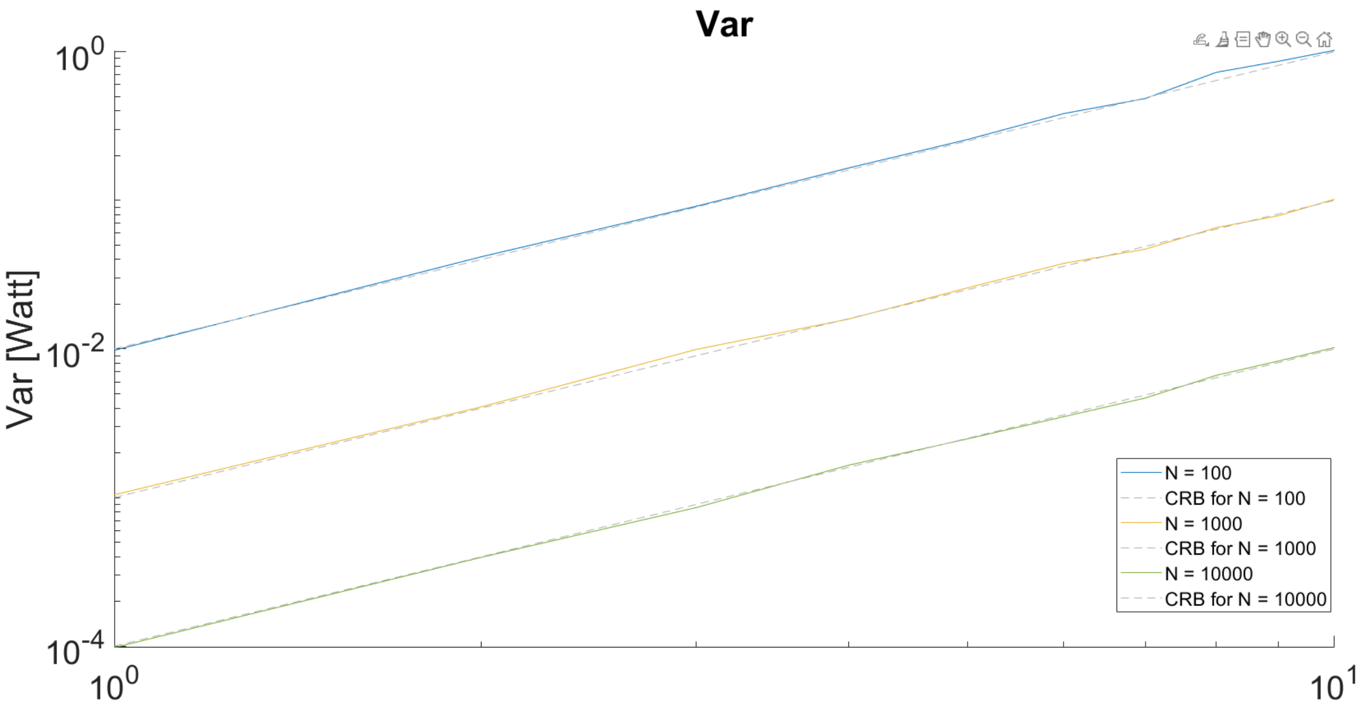


Plotted as a function of θ :

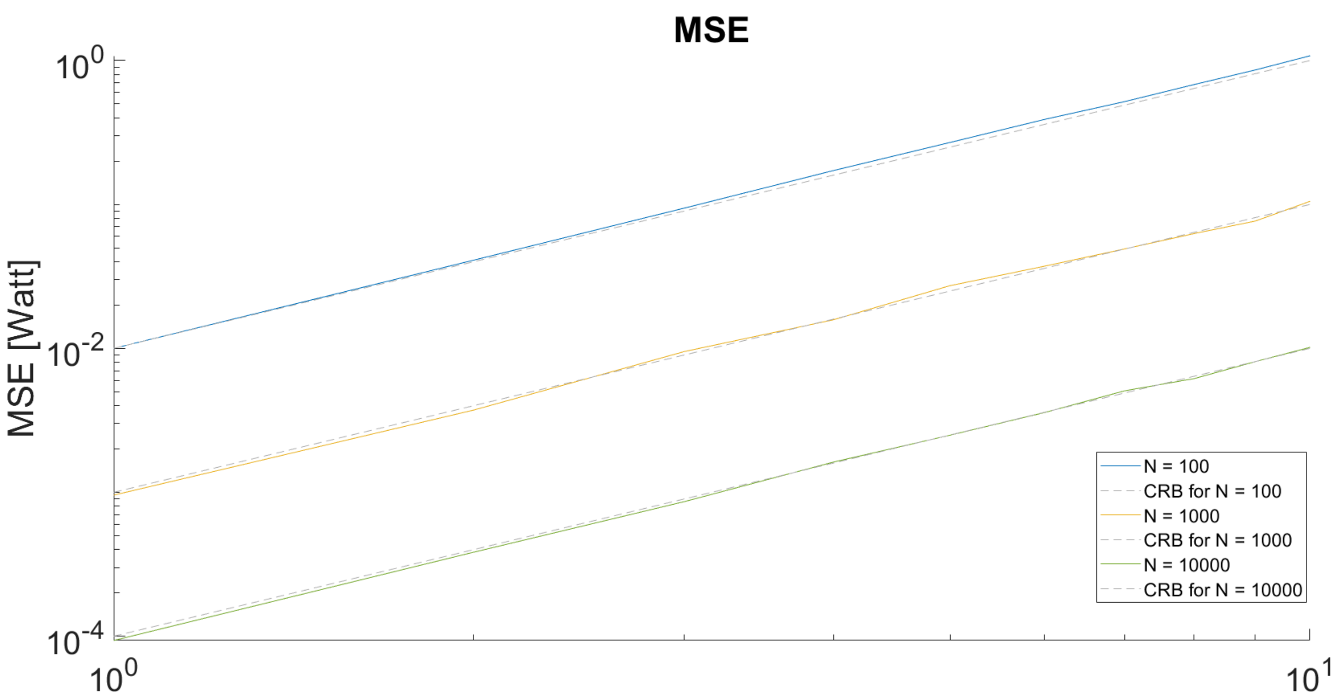
$Bias(N_i, \theta)$:



$Var(N_i, \theta)$:



$MSE(N_i, \theta)$:



For analysis purposes, we want to check if our estimator is unbiased and therefore to know if the CRLB would even possibly apply to our problem.

$$Bias(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}_{\underline{x}}[\hat{\theta}] - \theta$$

- To evaluate the following expected value we will use the MGF of a GWN vector :

$$\implies \mathbb{E}[\hat{\theta}] = \mathbb{E}[e^{\frac{1}{N}\sum_{i=1}^{N} x_i}] = \prod_{i=1}^{N} \mathbb{E}[e^{\frac{x_i}{N}}] = \prod_{i=1}^{N} M_{x_i}(t)|_{t=\frac{1}{N}} = \prod_{i=1}^{N} e^{(t\cdot\mu_i + t^2\cdot\sigma_i^2)}\Big|_{t=\frac{1}{N}} =$$

$$= \prod_{i=1}^{N} e^{\left(\frac{1}{N}\cdot\log\theta + \frac{1}{N^2}\cdot 1\right)} = e^{\log\theta} \cdot e^{\frac{1}{N}} = \theta \cdot e^{\frac{1}{N}}$$

- We plug it back in the expression for the bias :

$$\implies Bias(\hat{\theta}, \theta) = \theta \cdot e^{\frac{1}{N}} - \theta = \theta \cdot (e^{\frac{1}{N}} - 1)$$

✓ **Result**

We get a final expression for the bias of the estimator and we notice that it is **Asymptotically Unbiased :**

$$Bias(\hat{\theta}, \theta) = \theta \cdot (e^{\frac{1}{N}} - 1) \xrightarrow{N\to\infty} 0$$

- We will calculate the CRLB from the definition :

$$Var(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1} = CRLB$$

- We will derive the Fisher-Information from the definition. Since there are N i.id. samples, we can take the Fisher-Information of a single sample and multiply it by N :

$$\forall n, m \in \{1 \dots N\} \; : \; x_n \perp\!\!\!\perp x_m \implies \mathcal{I}(\theta) = N \cdot \mathbb{E}\left[\left(\frac{\partial \log f_{x_n|\theta}(x_n;\theta)}{\partial\theta}\right)^2\right]$$

$$\log f_{x_n|\theta}(x_n;\theta) = \log\left[(2\pi)^{-\frac{1}{2}} \cdot \exp\left(-\frac{(x_n - \log\theta)^2}{2}\right)\right] = -\frac{1}{2}\log(2\pi) - \frac{(x_n - \log\theta)^2}{2}$$

$$\xrightarrow{\frac{\partial}{\partial\theta}} \cancel{-}\frac{1}{\cancel{2}}(x_n - \log\theta) \cdot \cancel{2} \cdot (\cancel{-}\frac{1}{\theta}) = (x_n - \log\theta) \cdot \frac{1}{\theta}$$

$$N \cdot \mathbb{E}\left[\left(\frac{\partial \log f_{x_n|\theta}(x_n;\theta)}{\partial\theta}\right)^2\right] = N \cdot \mathbb{E}\left[(x_n - \log\theta)^2 \cdot \frac{1}{\theta^2}\middle|\theta\right] = N \cdot \frac{1}{\theta^2} \cdot \mathbb{E}[(x_n - \log\theta)^2]$$

- We can notice that by definition, the expected value in question is the Variance of the conditional distribution $x_n|\theta$ which is given to be 1 :

$$\dots = N \cdot \frac{1}{\theta^2} Var(x_n|\theta) = N \cdot \frac{1}{\theta^2}$$

- We get the final expression for the Fisher-Information :

$$\implies \mathcal{I}(\theta) = \frac{N}{\theta^2}$$

✓ **Result**

And finally, we get the final expression for the CRLB :

$$CRLB = \mathcal{I}(\theta)^{-1} = \frac{\theta^2}{N}$$

c) Efficiency

c. Is the proposed estimator efficient?

Next, we want to check the condition for which the CRLB can be attained. As we proved in class, the conditional PDF must satisfy :

$$\frac{\partial \log f_{\underline{x}|\theta}(\underline{x};\theta)}{\partial \theta} = c(\theta) \cdot (\varphi(x) - \theta)$$

- We will expand the terms for our problem (relying on earlier derivation...) to check if the condition holds :

$$\frac{\partial \log f_{\underline{x}|\theta}(\underline{x};\theta)}{\partial \theta} = \cdots = \underbrace{\frac{N}{\theta}}_{:=c(\theta)} \cdot \left[\underbrace{\frac{1}{N}\sum_{i=1}^{N} x_i}_{:=\varphi(x)} - \underbrace{\log \theta}_{\neq \theta} \right]$$

- There is no possible way to bring the $\log \theta$ term to the form of θ without breaking the definition for the rest of the terms...

× **Result**

We notice that we are not able to bring the score of the conditional distribution to the desired form and therefore we conclude that the CRLB is **unattainable** for our problem and in particular, to our estimator.

$$\implies Var(\hat{\theta}) > \mathcal{I}(\theta)^{-1} = CRLB$$

d) Efficiency

d. Is the proposed estimator asymptotically efficient? Your answer should be based both on simulation results and on theoretical analysis.

- ...However, the estimator I proposed is only **Asymptotically Unbiased**. That means that for any finite sample vector, no matter how large it is, the estimator is biased and the **CRLB** does not apply to it.
- It is visible from our plots that this projection holds since the bias always exist even though it is asymp. zero. The variance attains the CRB for each plot, matching our earlier projection. The MSE follows the same trend as the variance since the bias squared term goes to zero and the MSE and the variance align with each other.

Question 2

2. Let $x_1,\dots,x_N$ be statistically independent Gaussian random variables with mean μ and standard deviation σ.

We are given the statistical model of the data samples :

$$\underline{x} := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : \forall i \neq j \in \{1,\dots,n\} x_i \perp\!\!\!\perp x_j$$

$$\implies \underline{x} \sim \mathcal{N}(\mathbb{1}\cdot\mu\ ,\ \sigma^2 \cdot \mathbb{I}_n)$$

a) Find CRB

a. Find the CRB for estimation $\left[\mu,\sigma^2\right]^T$. Is there any coupling between the two

We want to find the CRLB for the case of a GWN vector.

- We will find it based on the definition :

$$\underline{\theta} := [\mu,\sigma]^T \implies \left[\underline{\underline{\mathcal{I}}}(\underline{\theta})\right]_{i,j} = \mathbb{E}\left[\frac{\partial \log f_{\underline{x}|\underline{\theta}}(\underline{x})}{\partial\theta_i}\cdot\frac{\partial \log f_{\underline{x}|\underline{\theta}}(\underline{x})}{\partial\theta_j}\right]$$

$$\log f_{\underline{x}|\underline{\theta}}(\underline{x}) = -\frac{n}{2}(\log(2\pi)+\log(\sigma^2)) - \sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}$$

$$\xrightarrow{\frac{d}{d\mu}} \quad \sum_{i=1}^{n}\frac{(x_i-\mu)}{\sigma^2}$$

$$\xrightarrow{\frac{d}{d\sigma^2}} \quad -\frac{n}{2\sigma^2}+\frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2$$

- Calculating the off-diagonal terms :

$$[\mathcal{I}(\underline{\theta})]_{1,2} = [\mathcal{I}(\underline{\theta})]_{2,1} = \mathbb{E}\left[\left[-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2\right]\cdot\sum_{i=1}^{n}(x_i-\mu)\right] =$$

$$= \frac{1}{2\sigma^2}\mathbb{E}\left[n\sum_{i=1}^{n}\underbrace{\cancel{(x_i-\mu)}}_{x_i\sim\mathcal{N}(\mu,\sigma^2)} + \sum_{i=1}^{n}\frac{(x_i-\mu)^3}{\sigma^2}\right] = \frac{1}{2\sigma^4}\sum_{i=1}^{n}\mathbb{E}\left[(x_i-\mu)^3\right] =$$

- To evaluate the expected value in question we will use the MGF of a joint-gaussian random vector :

$$\underline{z} := (\underline{x} - \mathbb{1}\cdot\mu) \sim \mathcal{N}(\underline{0}\ ,\ \sigma^2\cdot\mathbb{I}_n) \implies \mathbb{E}[z_i^n] = \begin{cases} 0 & : even \\ \sigma^n\cdot(n-1)!! & : odd \end{cases}$$

$$\implies \mathbb{E}[(x_i-\mu)^3] = 0 = [\mathcal{I}(\underline{\theta})]_{1,2} = [\mathcal{I}(\underline{\theta})]_{2,1}$$

- Calculating the diagonal terms :

$$[\mathcal{I}(\underline{\theta})]_{1,1} = -\mathbb{E}\left[\frac{d}{d\mu}\left(\sum_{i=1}^{n}\frac{(x_i-\mu)}{\sigma^2}\right)\right] = \sum_{i=1}^{n}\frac{1}{\sigma^2} = \frac{n}{\sigma^2}$$

$$[\mathcal{I}(\underline{\theta})]_{2,2} = -\mathbb{E}\left[\frac{d}{d(\sigma^2)}\left(-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2\right)\right] = -\mathbb{E}\left[\frac{n}{2\sigma^4} - \frac{1}{\sigma^6}\sum_{i=1}^{n}(x_i-\mu)^2\right] =$$

$$= -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6}\sum_{i=1}^{n}\mathbb{E}\left[(x_i-\mu)^2\right] = -\frac{n}{2\sigma^4} + \frac{n\cdot Var(x_i)}{\sigma^6} = -\frac{n}{2\sigma^4} + \frac{n\cdot\sigma^2}{\sigma^6} = \frac{n}{2\sigma^4}$$

- We get the FIM of the data samples :

$$\mathcal{I}(\underline{\theta}) = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}$$

✓ Result

We get the final expression for the CRB matrix :

$$\mathcal{I}(\underline{\theta})^{-1} = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix}$$